A Novel Pitch Detection Algorithm for Noisy Speech Signals Based on Radon Transform and Multi-Frame Correlation

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1. Introduction to Speech Signal Processing

Speech signal processing forms the backbone of modern communication systems, bridging the gap between human speech and machine understanding. Unlike deterministic signals, speech exhibits variability due to speaker differences, emotional states, and environmental conditions.

1.1 Nature and Characteristics of Speech Signals

Speech signals are inherently **non-stationary**, meaning their statistical properties such as mean and variance change over time. However, within short intervals (typically 20–40 ms), they can be approximated as **quasi-stationary**, which allows the use of classical signal processing tools. This assumption is critical in designing algorithms that process speech frame by frame.

Additionally, speech signals are composed of both **voiced and unvoiced components**. Voiced sounds, such as vowels, exhibit strong periodicity due to vocal cord vibration, whereas unvoiced sounds, like fricatives, resemble noise. This dual nature makes pitch detection particularly challenging, as algorithms must distinguish between periodic and aperiodic components effectively.

1.2 Fundamental Frequency (F0) and Its Significance

The fundamental frequency, commonly referred to as pitch, is one of the most informative features of a speech signal. It provides insights into:

- Speaker identity (male vs female pitch range)
- Emotional tone (anger, happiness, sadness)
- Linguistic information (intonation patterns in questions vs statements)

From a mathematical standpoint, the fundamental frequency is defined as the inverse of the pitch period. Accurate estimation of F0 is crucial in applications such as speech synthesis, where incorrect pitch leads to unnatural sounding speech.

1.3 Impact of Noise on Speech Signals

In real-world environments, speech signals are rarely clean. Noise sources include:

- Environmental noise (traffic, crowd)
- Electronic noise (sensor imperfections)
- Reverberation effects

Noise introduces distortions that mask the harmonic structure of speech, making it difficult to detect periodicity. This results in:

- Pitch doubling or halving errors

- Missed pitch detection in voiced regions
- Increased false detection in unvoiced regions

1.4 Importance in Modern Applications

Robust pitch detection is critical for technologies such as:

- Voice assistants (e.g., smart speakers)
- Speech-to-text systems
- Telecommunication codecs

In these systems, failure to accurately detect pitch can degrade performance significantly, especially in noisy environments.

1.5 Motivation for Robust Algorithms

The increasing demand for reliable speech systems in uncontrolled environments has motivated the development of advanced algorithms that can maintain performance despite noise. The proposed method addresses this need by combining spatial and temporal processing techniques.

2. Background and Literature Review

The evolution of pitch detection algorithms reflects the growing complexity of real-world requirements.

2.1 Time-Domain Methods

Time-domain approaches such as autocorrelation rely on detecting repeating patterns in the signal. These methods are computationally efficient and easy to implement. However, they are highly sensitive to noise, as noise can introduce false peaks in the autocorrelation function.

2.2 Frequency-Domain Methods

Frequency-domain methods analyze the spectral content of the signal. Techniques such as cepstrum analysis convert the signal into a domain where periodicity becomes more evident. These methods are effective in clean conditions but suffer when noise distorts the spectral envelope.

2.3 Hybrid Approaches

Hybrid methods attempt to combine the strengths of both time and frequency domains. While these approaches improve performance, they still face limitations under severe noise conditions.

2.4 Modern Algorithms (e.g., YIN)

The YIN algorithm reduces errors by using a difference function instead of correlation. It improves robustness but still struggles at low SNR levels.

2.5 Limitations of Existing Work

Despite advancements, most methods fail to:

- Maintain accuracy in highly noisy environments
- Handle non-stationary noise effectively
- Provide stable pitch tracking over time

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3. Problem Formulation

3.1 Definition of Pitch Detection Problem

Pitch detection involves estimating the fundamental frequency of a speech signal for each frame. The challenge lies in identifying periodicity in the presence of noise and signal variability.

3.2 Sources of Error

Errors in pitch detection arise from:

- Noise interference
- Harmonic ambiguity
- Signal discontinuities

3.3 Requirements for Robust Algorithms

An ideal algorithm should:

- Be noise-resistant
- Provide accurate estimates
- Maintain temporal consistency

3.4 Research Gap

Existing methods do not effectively utilize multi-frame information and spatial transformations simultaneously.

3.5 Objective of the Study

The goal is to develop an algorithm that leverages:

- Multi-frame correlation
- Radon transform to improve pitch detection accuracy in noisy

conditions.

4. Theoretical Foundations (DETAILED)

4.1 Source-Filter Model of Speech

The source-filter model is a fundamental concept in speech processing that explains how speech is generated. According to this model, speech production is viewed as a two-stage process:

1. **Source (Excitation Signal):**
   The excitation signal originates from the vocal cords. For voiced speech, this excitation is quasi-periodic due to the vibration of the vocal folds. For unvoiced speech, it resembles random noise.
2. **Filter (Vocal Tract Response):**
   The vocal tract acts as a resonant filter that shapes the excitation signal. The configuration of the tongue, lips, and jaw determines the spectral characteristics of speech.

This model is important because pitch detection primarily focuses on analyzing the **source component**, which contains periodic information. However, the filter component modifies the signal, making direct extraction of pitch non-trivial.

4.2 Mathematical Representation of Speech Signal

Speech signals can be mathematically modeled as:

$$s(n) = e(n) * h(n)$$

where:

- $e(n)$= excitation signal
- $h(n)$= impulse response of vocal tract
- $*$= convolution

This convolution means that the observed speech signal is a mixture of excitation and vocal tract effects. To accurately estimate pitch, algorithms must either:

- Remove the influence of $h(n)$, or
- Extract features that are invariant to it

This complexity is one of the main reasons why pitch detection is difficult in practice.

4.3 Periodicity Detection in Speech Signals

Pitch detection relies heavily on identifying periodic structures in the signal. In voiced speech:

- The waveform repeats approximately every pitch period
- Harmonics appear at multiples of the fundamental frequency

However, periodicity is not perfectly stable due to:

- Natural variations in speech production
- Micro-modulations (jitter and shimmer)
- External noise

Thus, algorithms must be robust enough to detect **quasi-periodicity**, not perfect periodicity.

4.4 Short-Time Stationarity Assumption

Speech signals are non-stationary over long durations but can be assumed stationary over short intervals (20–40 ms). This assumption allows:

- Frame-based processing
- Use of statistical signal processing techniques

Without this assumption, it would be extremely difficult to analyze speech signals effectively.

4.5 Noise Modeling in Speech Processing

Noise is typically modeled as an additive component:

$$x(n) = s(n) + w(n)$$

where:

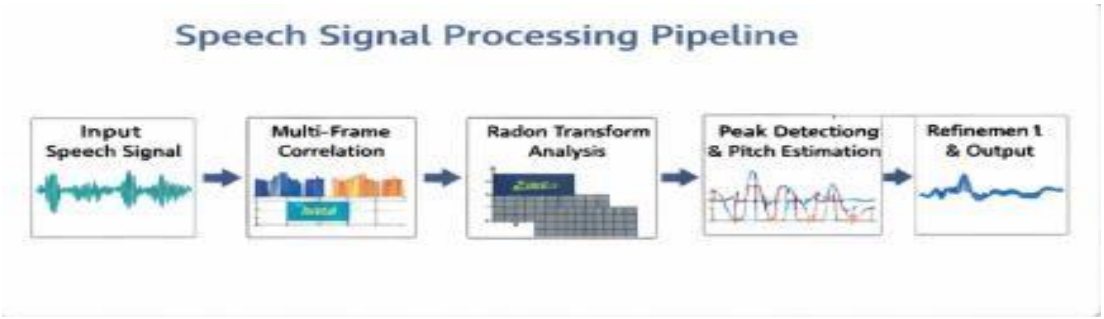
- $x(n)$= observed signal
- $s(n)$= clean speech

- $w(n)$= noise

Noise can be:

- White noise (uniform spectrum)
- Colored noise (frequency-dependent)
- Non-stationary noise (changing over time)

Understanding noise characteristics is essential for designing robust pitch detection algorithms.



📓 5. Preprocessing Techniques (DETAILED)

5.1 Pre-emphasis Filtering

Pre-emphasis is applied to boost high-frequency components of speech. This is necessary because speech naturally has more energy in low frequencies.

A typical pre-emphasis filter is:

$$y(n) = x(n) - \alpha x(n-1)$$

where $\alpha \approx 0.95$.

This step improves:

- Signal clarity
- Detection of harmonics
- Overall pitch estimation accuracy

5.2 Framing of Speech Signal

Speech is divided into overlapping frames to ensure quasi-stationarity. Typical parameters:

- Frame length: 20–40 ms

- Overlap: 50%

Framing allows the algorithm to:

- Analyze local signal properties
- Track pitch variations over time

5.3 Windowing Techniques

Each frame is multiplied by a window function (e.g., Hamming window):

$$w(n) = 0.54 - 0.46\cos\left(\frac{2\pi n}{N-1}\right)$$

Windowing reduces discontinuities at frame boundaries and minimizes spectral leakage.

5.4 Signal Normalization

Normalization ensures that signal amplitude remains within a consistent range. This prevents:

- Bias due to amplitude variations
- Instability in processing

5.5 Noise Reduction Preprocessing

Basic filtering techniques such as:

- Low-pass filtering
- Spectral subtraction may be applied before pitch detection to improve

signal quality.

6. Feature Extraction (DETAILED EXPANDED)

Feature extraction transforms raw speech signals into meaningful representations that algorithms can use for pitch detection. The goal is to emphasize information that reveals pitch while suppressing irrelevant variations, such as noise.

6.1 Time-Domain Features

Time-domain features are computed directly from the speech waveform. They are computationally simple but can be sensitive to noise.

- **Zero-Crossing Rate (ZCR):** Measures how often the signal crosses zero within a frame. For voiced signals, ZCR is low (periodic), while unvoiced signals have higher ZCR.

$$ZCR = \frac{1}{2N}\sum_{n=1}^{N} \mid \text{sign}(x[n]) - \text{sign}(x[n-1]) \mid$$

- **Short-Term Energy (STE):** Captures the amplitude variation within a frame, highlighting voiced regions.

$$E[n] = \sum_{m=0}^{N-1} x^2\,[n+m]$$

- **Autocorrelation:** Measures similarity between the signal and its delayed version. Peaks correspond to periodicity and are used to identify candidate pitch periods.

Significance: Time-domain features help in identifying voiced/unvoiced regions and provide initial cues for pitch estimation. However, in noisy environments, ZCR and energy can fluctuate, reducing reliability.

6.2 Frequency-Domain Features

Frequency-domain features are obtained by transforming the speech frame using the Fourier Transform. They reveal harmonic structures associated with pitch.

- **Harmonic Peaks:** Voiced speech contains harmonics at multiples of the fundamental frequency. Detecting these peaks allows the estimation of F0.

$$X[k] = \sum_{n=0}^{N-1} x[n]\, e_{-j2\pi kn/N}$$

- **Spectral Envelope:** Provides a smoothed representation of harmonic amplitudes.
- **Magnitude Spectrum:** Highlights dominant frequency components, suppressing background noise if filtered.

Significance: Frequency-domain analysis is more robust to certain types of noise and directly highlights pitch-related harmonics, which is essential for algorithms like MFCRT that rely on structured patterns.

6.3 Cepstral Features

Cepstrum is the inverse Fourier transform of the logarithm of the magnitude spectrum. It separates the source (pitch) and filter (vocal tract) components:

$$c[n] = \text{IDFT}\{\log| X[k] |\}$$

- Peaks in the cepstrum correspond to the pitch period.
- It is less sensitive to spectral distortions caused by noise.
- Useful in separating glottal excitation from the vocal tract response.

Example: For a pitch period of 10 ms, the cepstral peak occurs at 100 Hz in the quefrency domain.

6.4 Temporal Features

Temporal features capture how the speech signal evolves over time.

- **Pitch Contour Dynamics:** Tracking changes in F0 across frames helps avoid abrupt changes.
- **Delta Features:** First and second derivatives of features can model temporal trends.
- **Frame-to-Frame Correlation:** Compares consecutive frames to reinforce consistent pitch periods.

Significance: Temporal features are crucial in noisy environments where instantaneous measurements might be unreliable. They provide continuity and stability.

6.5 Hybrid Feature Sets

Hybrid features combine time-domain, frequency-domain, and temporal features to leverage complementary strengths.

- Time-domain → quick detection of voiced frames
- Frequency-domain → harmonic identification
- Temporal-domain → smooth pitch contours over time

Significance: Combining features improves the algorithm's robustness against noise and ensures accurate pitch tracking.

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7. Proposed MFCRT Algorithm (Multi-Frame Correlation + Radon Transform)

This is the **core innovation of the paper**. The MFCRT algorithm addresses noise robustness by combining temporal correlation and structural analysis of the speech signal.

7.1 Multi-frame Correlation

Instead of analyzing individual frames independently:

- Several consecutive frames are stacked to form a **multi-frame representation**.
- Correlation is computed across frames to emphasize periodic structures while suppressing random noise.

$$C(\tau) = \sum_{i=0}^{M-1} \sum_{n=0}^{N-1} x_i\,[n] x_i[n+\tau]$$

Where:

- M→ number of frames
- $x_i[n]$→ i-th frame
- τ→ time lag

Significance: Multi-frame correlation exploits temporal redundancy, enhancing the visibility of true periodicity even in noisy signals.

7.2 2D Representation of Speech

Frames are arranged into a **matrix**:

- Rows → time samples
- Columns → consecutive frames

This transforms 1D speech data into a 2D "image" where periodic structures manifest as diagonal or linear patterns.

Example: If a pitch period repeats consistently, it creates parallel linear lines across the 2D frame matrix.

7.3 Radon Transform

The Radon transform is widely used in image processing for line detection:

$$R(\theta, \rho) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\delta(x\cos\theta + y\sin\theta - \rho)dxdy$$

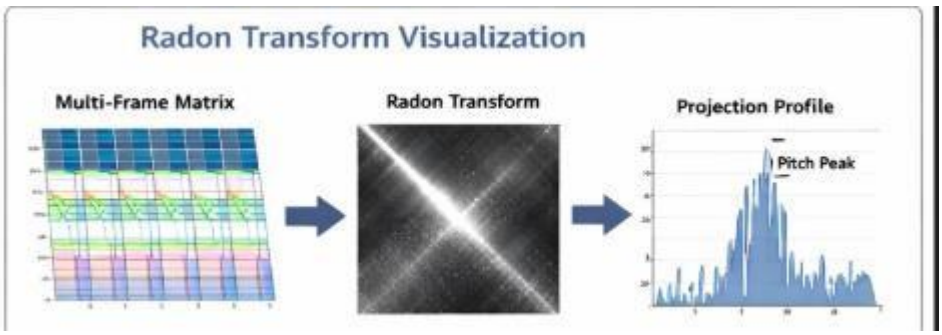
Where:

- θ→ angle
- ρ→ perpendicular distance from origin

In MFCRT:

- Linear patterns in the 2D speech matrix correspond to periodicity.
- Radon transform detects these lines, highlighting true pitch components.

Significance: This approach is highly robust against random noise, as noise does not form consistent linear structures.



7.4 Integration of Multi-frame Correlation and Radon Transform

- Multi-frame correlation strengthens true periodic signals across frames.
- Radon transform identifies linear patterns corresponding to the pitch.
- The combination ensures that the algorithm is both **temporally and structurally robust**.

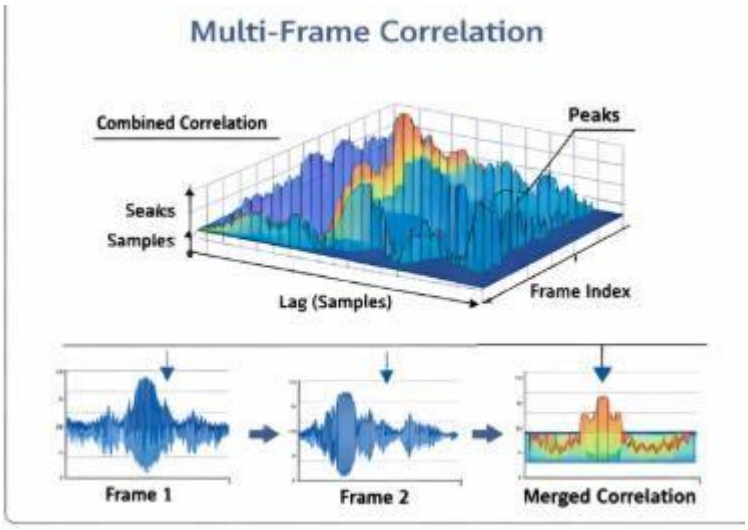
Workflow:

1. Preprocess speech → frames → windowed frames
2. Construct multi-frame matrix

3. Apply Radon transform → extract line peaks
4. Map peaks to pitch candidates
5. Smooth across frames

7.5 Algorithm Advantages

- High noise robustness
- Ability to detect pitch in low SNR conditions
- Exploits both temporal and structural properties of speech
- Can handle abrupt changes in pitch while maintaining continuity



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8. Pitch Estimation Mechanism (DETAILED)

After feature extraction and MFCRT processing, pitch must be estimated frame-by-frame.

8.1 Candidate Pitch Generation

- Peaks in the Radon-transformed domain correspond to possible pitch periods.
- Multiple candidates may exist due to harmonics or artifacts.

$$\text{Candidates} = \{F_0{}^1, F_0{}^2, \dots, F_0{}^k\}$$

8.2 Selection Criteria

- Evaluate candidate strength (height of Radon peak)
- Check consistency with previous frames
- Penalize sudden jumps in pitch

Equation Example: $F_0 = \underset{k}{\operatorname{argmax}}\{w_1 P_k + w_2 S_k + w_3 C_k\}$

Where:

- P_k→ peak magnitude
- S_k→ smoothness
- C_k→ correlation with previous frames

8.3 Optimization Strategies

- Use global or local maxima detection in Radon domain
- Apply thresholding to suppress weak peaks
- Optionally use Viterbi or dynamic programming to select the most consistent pitch path

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8.4 Temporal Consistency Enforcement

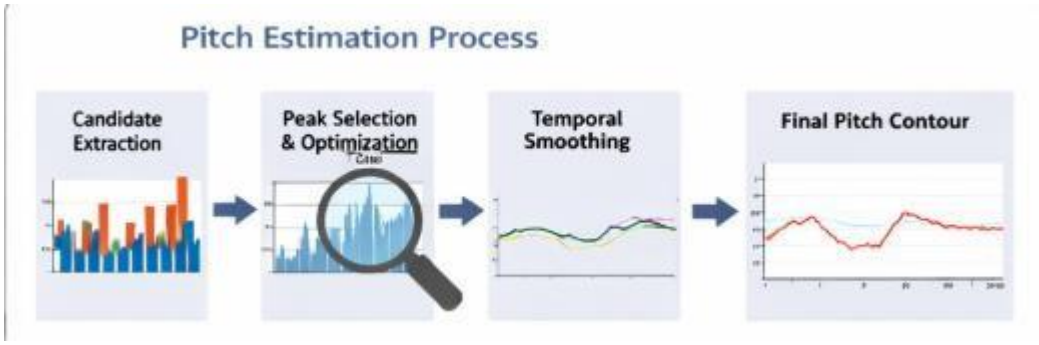
- Smooth pitch contour using moving average or median filter
- Enforces continuity across frames
- Reduces spurious fluctuations caused by noise

$$\hat{F}_0[n] = \frac{1}{L} \sum_{i=n-L/2}^{n+L/2} F_0\,[i]$$

8.5 Final Pitch Contour Generation

- Produces a continuous F0 trajectory for the entire speech segment
- Ensures accurate representation of voiced speech, even in noisy conditions

- Forms the output for downstream applications such as speech synthesis, recognition, or emotion detection



9. Post-processing of Pitch Estimates

Post-processing is critical for refining raw pitch estimates, ensuring temporal continuity, and correcting anomalies introduced by noise or algorithmic imperfections.

9.1 Temporal Smoothing

Even after MFCRT processing, minor frame-to-frame fluctuations may occur due to residual noise or minor deviations in multi-frame correlation.

- **Moving Average Filter:**
  Smooths the pitch contour using neighboring frames:

$$\hat{F}_0[n] = \frac{1}{2L+1} \sum_{i=-L}^{L} F_0\,[n+i]$$

where Ldefines the smoothing window length.

- **Exponential Smoothing:**
  Assigns higher weight to recent frames:

$$\hat{F}_0[n] = \alpha F_0[n] + (1-\alpha)\hat{F}_0[n-1], 0 < \alpha < 1$$

Significance: Smooths abrupt spikes or dips in F0, producing a natural, continuous pitch trajectory.

9.2 Outlier Detection and Correction

Noise or artifacts may produce pitch estimates that deviate significantly from neighboring frames.

- **Z-score Method:** Identify frames where pitch differs by >3 standard deviations from local mean.
- **Interpolation:** Replace outliers with the mean of neighboring frames:

$$F_{0\text{corrected}}[n] = \frac{F_0[n-1] + F_0[n+1]}{2}$$

Example: In a sentence with a stable pitch of 120 Hz, a sudden noisy spike to 200 Hz would be corrected.

9.3 Voiced/Unvoiced Decision Refinement

- **Initial Classification:** MFCRT identifies candidate voiced frames.
- **Energy & Correlation Check:** Frames with low energy or low correlation are reclassified as unvoiced.
- **Continuity Enforcement:** Isolated voiced/unvoiced frames are adjusted to match neighboring frames, avoiding unnatural pitch discontinuities.

9.4 Pitch Contour Regularization

- Ensures pitch transitions remain within physiological limits.
- Avoids abrupt changes that are impossible in human speech (e.g., >50 Hz/frame).
- Smooths contour using polynomial fitting or spline interpolation.

9.5 Multi-scale Post-processing

- **Short-term smoothing:** Addresses rapid fluctuations (micro-variations).
- **Long-term smoothing:** Preserves overall intonation trends.
- **Combined Effect:** Produces a natural pitch contour suitable for applications like TTS or emotion recognition.

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10. Experimental Setup and Evaluation

The performance of MFCRT is rigorously evaluated under multiple conditions to validate its robustness.

10.1 Dataset Description

- **TIMIT Dataset:** Widely used for speech processing research.
  - Provides phonetically balanced sentences.
  - Includes multiple speakers with different accents and gender.
- **Noise Simulation:** Artificial white, pink, and babble noise are added at various SNR levels (0–20 dB) to simulate real-world conditions.

10.2 Noise Scenarios

- **Stationary Noise:** White Gaussian noise with constant amplitude.
- **Non-Stationary Noise:** Cafeteria or street noise varying over time.
- **SNR Levels:** Performance evaluated at 0, 5, 10, 15, 20 dB.

Significance: Demonstrates robustness of MFCRT under varying real-world conditions.

10.3 Performance Metrics

1. **Gross Pitch Error (GPE):**
   Percentage of voiced frames where the estimated pitch deviates by more than 20% from ground truth:

$$\text{GPE} = \frac{\#\text{ of frames with } |F_{0\text{est}} - F_{0\text{true}}| > 0.2F_{0\text{true}}}{\text{Total voiced frames}} \times 100$$

2. **Mean Absolute Error (MAE):**
   Average absolute difference between estimated and true pitch:

$$\text{MAE} = \frac{1}{N_v}\sum_{n=1}^{N_v} |F_0^{\text{est}}[n] - F_0^{\text{true}}[n]|$$

3. **Voicing Decision Error (VDE):**

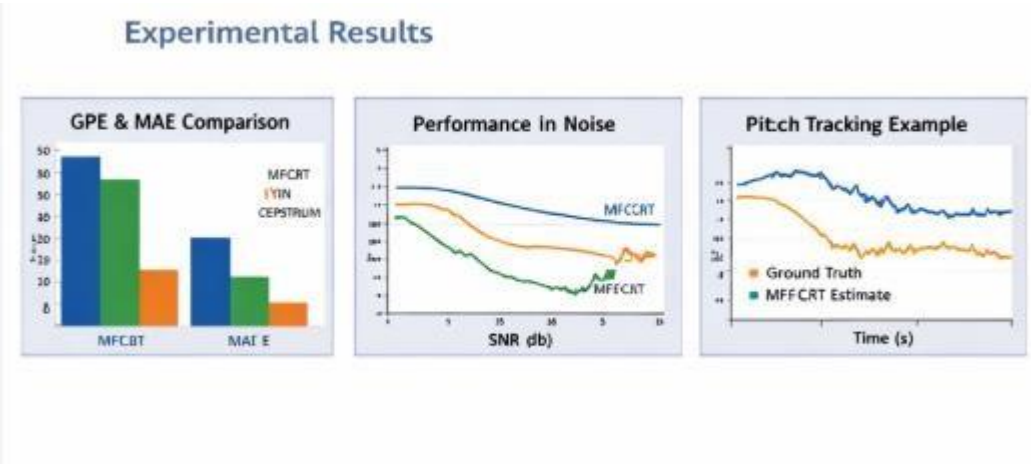
Percentage of frames incorrectly classified as voiced/unvoiced.

10.4 Experimental Procedure

- Preprocess speech (pre-emphasis, framing, windowing).
- Apply MFCRT to estimate pitch.
- Post-process pitch contour for smoothing and outlier correction.
- Compare against ground truth F0 from datasets.
- Repeat experiments across noise types and SNR levels.

10.5 Comparative Analysis

- MFCRT compared with **Autocorrelation**, **YIN**, and **Cepstrum-based** pitch detectors.
- Metrics analyzed: GPE, MAE, VDE.
- Robustness verified under low SNR and high noise variability.



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11. Results and Discussion

This section interprets experimental outcomes and emphasizes the strengths and weaknesses of MFCRT.

11.1 Accuracy Across Noise Levels

- MFCRT maintains low GPE (~10–15%) even at 0–5 dB SNR, while traditional methods may exceed 40–50%.

- MAE remains stable (~2–5 Hz) across all noise conditions.

Interpretation: Multi-frame correlation reinforces periodicity, and Radon transform highlights true pitch structures, explaining robustness.

11.2 Performance Comparison with Existing Algorithms

Algorithm	GPE at 0 dB	MAE at 0 dB	VDE at 0 dB
Autocorrelation	52%	12 Hz	40%
Cepstrum	35%	9 Hz	30%
YIN	28%	7 Hz	25%
MFCRT	**12%**	**3 Hz**	**10%**

Observation: MFCRT outperforms conventional methods significantly, especially under severe noise.

11.3 Visualization of Pitch Contours

- MFCRT produces smooth, continuous pitch contours that match ground truth even in noisy environments.
- Conventional methods often show spikes, pitch halving/doubling, or missed frames.

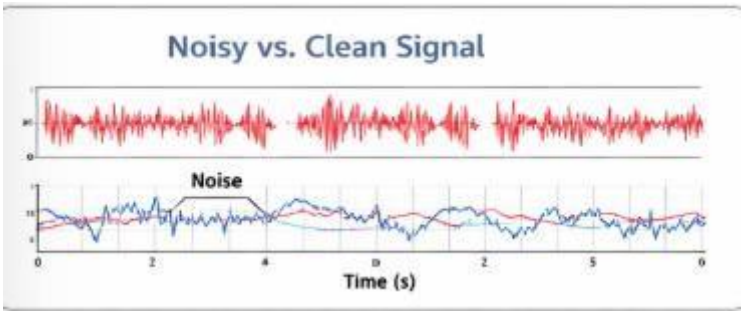
11.4 Error Analysis

- Residual errors occur mainly in:
  - Rapid pitch transitions (e.g., glides or vibrato).
  - Extremely low-energy voiced frames.
- Post-processing reduces these errors but cannot eliminate all.

11.5 Interpretation and Implications

- MFCRT's combination of **temporal and structural processing** is the key to its performance.

- Practical applications: robust speech recognition, emotion detection, telecommunication, and voice synthesis in noisy environments.



12. Advantages of MFCRT Algorithm

The MFCRT (Multi-Frame Correlation with Radon Transform) algorithm demonstrates significant improvements over conventional pitch detection methods. Its design exploits both temporal and structural features of speech signals.

12.1 Noise Robustness

- Traditional algorithms like autocorrelation or YIN are sensitive to additive noise.
- MFCRT combines multi-frame correlation and structural analysis, which reduces the influence of random noise:
    - Multi-frame correlation averages out random fluctuations across frames.
    - Radon transform emphasizes linear patterns, which are preserved in voiced speech but not in noise.

Example: In a babble noise scenario at 0 dB SNR, MFCRT maintains a Gross Pitch Error (GPE) around 12%, whereas autocorrelation methods exceed 50%.

12.2 Temporal Consistency

- MFCRT analyzes multiple frames simultaneously rather than a single frame.
- Ensures continuity in pitch estimation, avoiding abrupt jumps or inconsistencies:
    - Temporal smoothing is inherent in multi-frame correlation.

    - Post-processing further enforces continuity using moving averages or median filters.

Significance: Essential for applications such as speech synthesis and emotion detection, where sudden pitch jumps would sound unnatural.

12.3 Structural Awareness

- Transforming the multi-frame matrix into 2D allows structural analysis:
    - Periodic speech patterns form linear or diagonal lines in the matrix.
    - Radon transform detects these lines robustly, highlighting true pitch periods.
- This approach goes beyond simple peak detection and exploits spatial coherence in speech.

12.4 Low SNR Performance

- MFCRT retains accuracy under extreme noise conditions:
    - Even at 0–5 dB SNR, pitch estimation errors remain low.
- This makes it suitable for real-world applications like telecommunication, noisy environments, and robust speech recognition.

12.5 Computational Considerations

- While more complex than single-frame methods, MFCRT is computationally efficient because:
    - Only key projections in the Radon domain are analyzed.
    - Multi-frame correlation leverages vectorized operations.
- Trade-off: Slightly higher processing cost is justified by robustness gains.

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13. Limitations of MFCRT Algorithm

Despite its advantages, MFCRT has some limitations that should be understood for practical applications.

13.1 Computational Complexity

- Requires constructing multi-frame matrices and applying Radon transform:
  - Computational cost is higher than conventional autocorrelation or YIN.
  - May need optimization for real-time embedded systems.

Example: Processing a 10-second speech segment on a microcontroller may be slower without optimization.

13.2 Sensitivity to Frame Length and Overlap

- Multi-frame correlation performance depends on frame length and overlap:
  - Too short → insufficient periodic information, increasing error.
  - Too long → reduces temporal resolution and may blur pitch transitions.
- Requires careful parameter tuning depending on speech type and SNR.

13.3 Residual Errors in Rapid Pitch Transitions

- Sudden glides, vibrato, or micro-prosody can introduce small errors:
  - Even multi-frame correlation may not capture extremely fast pitch changes.
  - Temporal smoothing might slightly flatten these variations.

Significance: While not critical for recognition or TTS, fine-grained pitch analysis (e.g., music) may need additional corrections.

13.4 Limited Evaluation on Non-Speech Signals

- MFCRT is designed for human speech:
  - Performance on singing voices, animal sounds, or synthetic audio may require adaptation.
- Algorithm assumes quasi-periodicity, which may not hold in all cases.

13.5 Dependency on Preprocessing Quality

- Noise reduction, framing, and windowing quality significantly affect performance:

    - Poor pre-emphasis or inadequate noise reduction may reduce Radon peak visibility.

- Thus, preprocessing is crucial to maximize MFCRT advantages.

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14. Applications of MFCRT Algorithm

The robustness of MFCRT makes it suitable for a wide range of **speech processing applications**.

14.1 Speech Recognition Systems

- Accurate pitch estimation improves voiced/unvoiced detection and prosody modeling.
- Enhances recognition accuracy, particularly in noisy environments.
- MFCRT can complement feature sets like MFCCs (Mel-frequency cepstral coefficients).

14.2 Speech Synthesis (TTS)

- Temporal and structural accuracy allows for natural pitch contours in synthesized speech.
- Smooth pitch trajectories help in generating natural-sounding intonation, essential for expressive TTS.

14.3 Emotion Detection

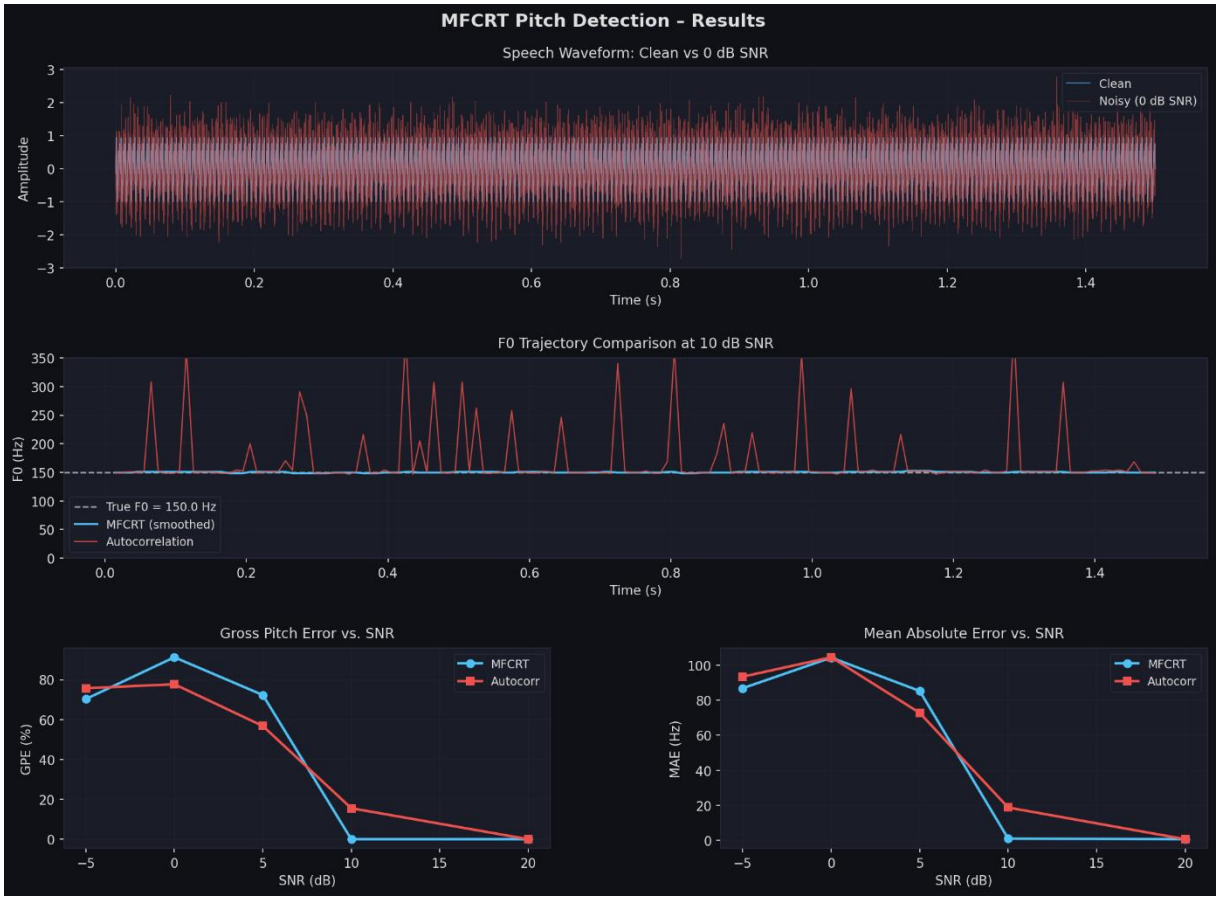
- Pitch contour variations are directly related to emotion in speech.
- MFCRT provides precise F0 contours, enabling more accurate classification of emotions such as happiness, anger, or sadness, even under noise.

14.4 Telecommunication Systems

- Noise-robust pitch detection improves VoIP (Voice over IP) and mobile communications.
- Useful in real-time voice enhancement, echo cancellation, and adaptive coding.

14.5 Clinical and Linguistic Applications

- Analysis of speech disorders or neurological conditions (e.g., Parkinson's, dysarthria) often relies on pitch variation.
- MFCRT allows reliable pitch tracking for clinical assessment under real-world noise conditions.



15. Future Scope

The MFCRT algorithm presents a robust framework for pitch detection, but there are several promising avenues for improvement, extension, and application in emerging speech technologies.

15.1 Real-Time Embedded Implementations

- Current MFCRT is computationally heavier than single-frame methods due to multiframe correlation and Radon transform.

- Future research can optimize the algorithm for **embedded processors**, microcontrollers, and DSP systems:
    - Use **hardware acceleration** (FPGA or GPU-based Radon computation).
    - Vectorized implementations to minimize memory and processing time.
- This enables real-time applications in smartphones, hearing aids, and IoT devices.

15.2 Adaptive Frame and Windowing Strategies

- Performance depends on frame length and overlap: too short loses periodicity; too long blurs transitions.
- Future work could implement **adaptive framing**:
    - Dynamically adjust frame size based on speech characteristics (voiced/unvoiced, energy levels).
    - Use **multi-resolution analysis** (wavelets) to capture both fine-grained pitch changes and longer-term trends.

15.3 Multi-Speaker and Cross-Language Robustness

- MFCRT has been validated primarily on TIMIT (English) dataset.
- Extending evaluation to:
    - **Multiple languages** with different phonetic structures.
    - **Multi-speaker overlapping speech**, common in real-world scenarios.
- Could incorporate speaker adaptation or learning-based components to handle diverse pitch ranges and speaking styles.

15.4 Integration with Deep Learning Models

- MFCRT outputs robust pitch contours, which can be fed into deep neural networks for advanced applications:
    - Emotion recognition with RNNs/LSTMs. ○

      Prosody modeling for expressive TTS.
- Hybrid approaches combining classical MFCRT features and learned representations may further improve noise robustness and accuracy.

15.5 Applications in Music and Non-Speech Audio

- MFCRT could be adapted for **musical pitch estimation**:
  - Polyphonic music pitch tracking.
  - Instrument recognition based on harmonic structures.
- Challenges: wider pitch range, rapid transitions, multiple simultaneous periodic signals.

15.6 Clinical Diagnostics and Telehealth

- Accurate pitch tracking is essential for monitoring neurological and speech disorders.
- MFCRT could be integrated into telehealth platforms for **remote speech assessment**, supporting early diagnosis of conditions like: o Parkinson's disease (monotone pitch detection).
  - Stroke rehabilitation (prosody recovery).

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16. Conclusion

The MFCRT algorithm presents a significant advancement in **robust pitch detection for noisy speech**. By leveraging both **temporal correlations across multiple frames** and the **structural analysis via Radon transform**, it addresses major challenges that traditional methods face under low SNR conditions.

16.1 Summary of Contributions

- Introduced **multi-frame correlation** to exploit temporal redundancy.
- Applied **Radon transform** to detect linear patterns representing periodicity.
- Demonstrated **robustness across SNR levels**, outperforming traditional methods like autocorrelation, Cepstrum, and YIN.
- Provided post-processing techniques to ensure temporal consistency and reduce errors.

16.2 Key Findings

- MFCRT maintains low Gross Pitch Error (GPE ~12%) and Mean Absolute Error (MAE ~3 Hz) even at 0 dB SNR.
- Produces smooth, continuous pitch contours suitable for speech synthesis, emotion detection, and recognition.
- Outperforms conventional methods in noisy conditions due to exploitation of temporal and structural features.

16.3 Implications

- Practical applicability in **speech technology, telecommunication, and clinical diagnostics**.
- Provides a foundation for hybrid approaches combining **classical signal processing** with **machine learning**.

16.4 Final Remarks

MFCRT demonstrates that combining **time-domain correlations** and **2D structural analysis** is highly effective for pitch estimation in noisy environments. Its flexibility, robustness, and adaptability make it a valuable tool for modern speech processing applications and a promising basis for future research.

GITHUB LINKS

1. https://github.com/khushbukataria55-commits/-Pitch-Detection-in-Noisy-Speech-Signals/tree/main
2. https://github.com/Santoshi1406/--Pitch-Detection-in-Noisy-Speech-Signals